

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	01 July 2026
Team ID	SWTID-2026-2152
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

1. Brainstorming

• Define the Problem:

Clearly identify the challenges in detecting human emotions from text, such as inaccurate emotion recognition, difficulty understanding user sentiment, lack of personalized responses, and limited emotional support in digital communication. The Emotion Detection Support Engine aims to address these challenges using Artificial Intelligence, Natural Language Processing (NLP), and Deep Learning techniques.

• Gather Participants:

Include project team members, mentor/guide, AI and NLP developers, and potential users to gather diverse ideas and perspectives for designing an effective emotion detection system.

• Use Brainstorming Techniques:

Mind Mapping: Organize ideas related to data collection, preprocessing, model development, AI response generation, user interface, and deployment.

Brainwriting: Team members individually suggest ideas for improving emotion detection accuracy, model selection, and system features before discussing them together.

SCAMPER: Apply techniques such as Substitute, Combine, Adapt, Modify, Put to other uses, Eliminate, and Reverse to generate innovative solutions and improve the overall system.

• Encourage Free Thinking:

Create an open environment where every team member can freely share ideas for improving emotion classification, AI-generated responses, dashboard design, and user experience without criticism.

• Document Everything:

Record all proposed ideas, workflows, and feature suggestions. These ideas are grouped and evaluated to identify the most suitable solution for implementation.

2. Idea Prioritization

• Establish Criteria:

Evaluate each idea based on:

Project Impact: How much will the idea improve emotion detection accuracy and user experience?

Feasibility: Can the idea be implemented using available datasets, technologies, and project resources?

User Benefit: Will the feature provide meaningful emotional insights and supportive responses?

Alignment with Project Goals: Does the idea contribute to building an intelligent and reliable Emotion Detection Support Engine?

Technical Complexity: How difficult is the feature to develop, test, and maintain?

• Prioritization Frameworks:

MoSCoW Method: Categorize features into Must Have, Should Have, Could Have, and Won't Have.

Impact/Effort Matrix: Prioritize features with high impact and low implementation effort.

Feature Ranking: Rank ideas according to project objectives, technical feasibility, and expected

user value.

- **Team Collaboration:**

The project team reviews and discusses all proposed ideas to ensure consensus on the most valuable features before implementation.

- **Communication and Review:**

The prioritized feature list is documented and reviewed regularly throughout the project to accommodate improvements and user feedback.

3. Implementation Planning

- **Develop a Roadmap:**

Create a phased implementation plan covering environment setup, data preprocessing, model training, emotion detection, AI response generation, Streamlit interface development, testing, and deployment.

- **Assign Responsibilities:**

Allocate responsibilities for data preparation, model development, UI design, testing, documentation, and deployment among the project team members.

- **Track Progress:**

Monitor project milestones using task management and regular progress reviews to ensure timely completion of each phase.

- **Monitor and Evaluate:**

Evaluate the system based on prediction accuracy, model performance, response quality, and overall user experience.

- **Gather User Feedback:**

Collect feedback from users to identify areas for improvement and enhance the accuracy, usability, and performance of the Emotion Detection Support Engine.

By following these brainstorming, idea prioritization, and implementation planning steps, the team developed a structured approach for building the Emotion Detection Support Engine. This process helped identify the most effective ideas, prioritize essential features, and create a clear development roadmap, resulting in an intelligent system capable of accurately detecting emotions from text and providing empathetic, AI-powered responses.

